

PIXE analysis on Maya blue in Prehispanic and colonial mural paintings

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Abstract

Particle induced X-ray emission (PIXE) experiments have been carried out at the AGLAE facility (Paris) on several mural samples containing Maya blue from different Prehispanic archaeological sites (Cacaxtla, El Tajín, Tamuin, Santa Cecilia Acatitlán) and from several colonial convents in the Mexican plateau (Jiutepec, Totimehuacán, Tezontepec and Cuauhtinchán). The analysis of the concentration of several elements permitted to extract some information on the technique used for painting the mural, usually fresco. Principal component analysis permitted to classify the samples into groups. This grouping is discussed in relation to geographic and historic data. © 2006 Elsevier B.V. All rights reserved.

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1. Introduction

Maya blue is an artificial pigment commonly used by Mayas and other people in Mesoamerica. It is made out from a fibrous clay mineral (palygorskite, with theoretical formula $\text{Si}_8\text{O}_{20}\text{Al}_2\text{Mg}_2(\text{OH})_2(\text{H}_2\text{O})_4 \cdot 4(\text{H}_2\text{O})$) plus an organic colorant (indigo: $\text{C}_{16}\text{H}_{10}\text{N}_2\text{O}_2$) [1,2]. It has light blue (or turquoise) color. It is found in murals, pottery, texts (codices), sculptures and ceremonial and funeral objects. It is believed that its use goes back to the VIII century AD and continued during colonization in convents until around 1580 [3]. Maya blue is also found in colonial paints in Cuba (XVIII–XIX) [4]. During the colonial period (1521–1821) native technologies were usually replaced by European technologies. However, it is interesting that most convents built during XVIth century were decorated

using mural paints, following a Prehispanic tradition, instead, or in addition to, the European tradition that preferred altarpieces. Moreover, these mural paints contain Prehispanic elements both in technology (they contain Maya blue, unknown in Europe) and in symbolism, as an example of the syncretism between the Prehispanic culture and christianity. In this work we performed particle induced X-ray emission (PIXE) analysis on mural samples containing Maya blue from different Prehispanic archaeological sites and colonial convents in the Mexican plateau, with the aim of evaluating differences and similarities between these two well defined groups. The analysis of the concentration of several major elements (e.g. Ca, Mg, Fe) and trace elements (e.g. Ni, Cu, Zn and Pb) was carried out in order to obtain information on the nature of the pigments, and gain information on the employed technologies. A previous work [5] reported some conclusions regarding the major elements Ca, Fe and Mg when comparing blue and red pigments with the plaster and substrate. In this

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work, we include recent measurements on more samples, including new Prehispanic areas (like Aztec one) and a convent from a different religious order (Agustinian). We now focus our study in the blue pigments from two well-defined groups of murals, Prehispanic and colonial.

2. Materials and methods

Archaeological samples were provided by one of the authors (Reyes-Valerio). Several samples were collected from five Mexican Prehispanic archaeological sites: Cacaxtla (Tlaxcala, c. X–XIII) [CA], Tamuín (San Luis Potosí, Huastec culture, c. XI, XII) [TAM], El Tajín (Veracruz, Totonaca culture, c. XI–XII) [TAJ] and Tajín Chico (Veracruz, c. XI–XII) [TAC] and Santa Cecilia Acatitlán (State of México, c. XIII–XVI, Aztec culture) [CEC]. Colonial samples were collected from Mexican Franciscan convents: Cuauhtinchán (Puebla, 1550–1570) [CUA], Totimehuacán (Puebla, XVI c.) [TOT] and Jiutepec (Morelos, 1540–1570) [JIU], and the Augustinian convent, Tezontepec (Hidalgo) [TEZ]. The collection sites are shown in Fig. 1. The samples from Jiutepec previously analyzed [5] were collected around 1980 in a convent chapel which is now completely destroyed. They contained a blue pigment which is not Maya blue, and come probably from a restoration. The Jiutepec samples analyzed in this work come from a decoration in the cloister, recently recovered, and they contain Maya blue (confirmed by its resistance to acids).

PIXE experiments were conducted in at the AGLAE (Accélérateur Grand Louvre d'Analyse Élémentaire) facility in Paris in April 2003 and in June 2004. The PIXE facility is based on a NEC 6SDH-2 2 MeV tandem Pelletron. A beam of about 100 μm was used and the sample was usually scanned in order to average over a $0.5 \times 0.5 \text{ mm}^2$ zone.

This scan also helped in minimizing the possible radiation damage on the pigment (no darkening of the irradiated surface was observed). Objects were set on a 3-D sample holder monitored by a video camera which allowed monitoring the exact point of impact of the proton beam, recognized by a luminescent point. Two Si(Li) detectors oriented at 45° to the proton beam recorded the X-ray spectra, being dedicated respectively, to the low-energy X-rays (0.3–10 keV) from low-Z major elements of the matrix, and the high energy X-rays (5–40 keV) emitted by trace elements. A 50 μm Al filter was used upstream from the high-energy detector, and the low-energy detector used a 8 μm Be window. Spectra were accumulated until an integrated dose of 250 000 counts was reached, which corresponds to about 5 min of acquisition per spectrum. Elemental concentration analysis was done using the GUPIX code [6,7] and calibration was done using the BE-N standard, a basalt rock [8]. Statistical analysis (correlation and multivariate analyses) has been performed using the STATISTICA software [9]. For the statistical analysis, we have used as input variables the logarithm (log 10) of the concentrations, to avoid great discrepancies in the concentration range.

3. Results and discussion

The resulting element concentrations from the quantification using GUPIX on a number of selected samples is shown in Table 1. The sample's labeling indicated with the first characters the provenance of the sample, the first digit means the sample number and the second one (after the underscore) means the number of measurement for a given sample. By looking at the concentration of some major elements, some conclusions already drawn in a previous publication [5] are confirmed on the new samples.



Fig. 1. Map of Mexico with the location of the different sites from where the studied samples were collected.

Table 1
Elemental concentration in percent (for the elements marked with asterisk) or $\mu\text{g/g}$ for the samples used in the PCA

	11-Na*	12-Mg*	13-Al*	14-Si*	16-S*	17-Cl*	19-K*	20-Ca*	26-Fe*	15-P	22-Ti	25-Mn	28-Ni	29-Cu	30-Zn	38-Sr	82-Pb
Jiu1_1	0.29	1.76	1.66	10.7	5.23	0.65	0.76	39.3	0.62	[0]	620	(120)	(15)	40	29	440	49
Tot1_1	0.84	1.74	1.85	5.49	2.17	0.35	1.64	35.1	0.61	99,600	840	(170)	(14)	27	164	420	59
Tot1_2	0.20	0.96	1.03	4.55	12.9	0.11	0.12	37.9	0.39	(1200)	810	(130)	(11)	(10)	21	1040	65
Tot1_3	0.45	1.62	2.97	11.6	4.75	0.38	1.71	26.4	1.64	51,800	5010	360	18	35	108	460	69
Tez1_1	0.29	0.68	1.26	5.58	9.89	0.45	0.10	41.6	0.38	[0]	540	(270)	(3)	(8)	(12)	1430	62
Tez1_2	0.32	0.74	1.47	6.57	8.67	0.51	0.15	41.6	0.55	[0]	660	(310)	(6)	(9)	15	1310	152
Tez2_1	1.26	2.79	1.91	7.00	1.83	1.32	0.18	48.8	0.39	[0]	600	(190)	(9)	19	(12)	900	49
Tez2_2	1.71	2.90	2.30	9.77	2.58	1.36	0.26	42.0	0.50	[0]	(520)	310	(14)	31	44	1080	49
Cua1_1	0.36	1.57	2.66	8.32	0.22	(0.03)	0.14	51.7	0.57	[0]	810	330	25	(14)	(14)	8580	21
Cua1_2	0.36	1.68	2.71	8.38	0.18	(0.02)	0.12	51.5	0.51	[0]	820	240	20	(10)	14	6550	1
Ca1_1	0.30	3.17	4.61	17.6	0.39	0.06	0.38	31.8	1.15	(170)	2070	330	26	22	18	560	15
Ca1_2	0.15	3.40	4.12	16.0	0.41	0.05	0.29	34.6	1.41	[0]	1980	630	32	(6)	34	430	2
CA2_1	0.51	3.03	4.52	16.4	0.90	0.08	0.34	33.1	0.99	[0]	1330	600	23	(4)	69	160	9
CA2_2	0.64	2.96	5.80	20.6	0.66	(0.07)	0.59	24.2	1.74	(430)	2340	550	17	(6)	16	100	2
CA2_3	(0.29)	3.19	4.51	16.3	0.43	(0.07)	0.32	34.2	0.75	[230]	2340	(260)	(9)	(2)	8	160	5
CA2_4	0.44	3.22	4.80	17.1	0.47	(0.05)	0.39	32.3	0.76	[270]	2310	(150)	14	(5)	14	210	7
CA3_1	0.49	1.39	3.98	11.6	0.51	0.09	0.29	43.5	0.93	(1300)	2140	(360)	16	(4)	14	160	4
CA3_2	0.70	1.37	3.62	11.7	1.00	(0.07)	0.60	42.2	0.97	(1700)	4560	(320)	(12)	(10)	23	270	7
Ca4_1	0.29	1.64	2.74	10.5	0.84	0.08	0.31	46.1	0.79	(940)	4580	380	22	(7)	25	1180	4
Ca4_2	0.33	1.46	3.08	11.2	0.52	0.07	0.24	45.5	0.78	(1260)	3300	580	18	(11)	26	850	18
Cec1_1	0.23	6.24	3.69	22.8	1.32	0.11	0.71	17.5	2.27	[0]	2110	9120	30	(12)	60	1170	7
Cec1_2	0.11	8.60	2.81	22.5	0.18	0.11	0.58	19.9	1.54	[0]	1820	4790	23	14	48	1260	18
Cec1_3	0.56	9.30	5.29	30.7	0.19	0.06	1.05	2.88	1.32	[0]	1320	1980	21	21	87	490	10
Cec1_4	0.22	8.40	4.54	29.6	0.32	0.06	1.01	6.07	1.54	[0]	1870	6730	26	21	88	810	17
Cec2_1	0.20	6.82	3.50	23.2	1.08	0.11	0.71	17.8	1.67	[0]	1600	8630	29	16	57	1190	13
Cec2_2	0.14	8.51	2.85	23.3	0.65	0.10	0.63	16.6	2.22	[0]	3880	7170	28	16	53	1300	13
TAM1_1	(0.29)	2.25	4.08	12.8	0.15	(0.02)	0.26	41.1	1.49	(850)	1370	(250)	15	(3)	12	500	6
TAM1_2	(0.19)	1.35	3.25	10.5	0.22	[0.01]	0.20	47.3	1.19	(850)	1040	(150)	(12)	(5)	(8)	300	5
TAJ1_1	1.49	0.67	7.43	16.5	(0.16)	(0.07)	0.43	31.7	1.38	[0]	1040	(280)	22	[0]	15	740	10
TAJ1_2	(0.20)	1.26	2.48	8.00	0.10	[0.01]	(0.06)	53.1	0.54	(1530)	850	(170)	(9)	(3)	10	90	3
TAJ1_3	(0.19)	3.37	4.90	17.9	(0.06)	(0.01)	0.27	31.9	0.76	(590)	1470	(140)	12	(5)	8	110	1
TAJ1_4	(0.43)	0.38	2.10	5.10	(0.09)	0.22	0.36	57.8	1.18	(690)	1600	(240)	(29)	(17)	21	570	1
TAC1_1	0.56	1.73	4.90	17.9	0.36	(0.02)	0.90	27.2	5.18	(1000)	3920	900	154	(12)	52	270	1
TAM1_1	0.40	0.98	3.95	10.2	0.14	[0.00]	0.14	47.4	0.86	(1320)	1310	(250)	(11)	(4)	12	320	1
TAM1_2	0.41	0.89	3.32	8.76	0.16	(0.01)	0.14	50.7	0.61	(1340)	1250	(200)	(14)	(3)	(6)	340	4

Numbers in squared brackets are close to the limit of detection (LOD), and numbers in brackets are below the LOD. Typical errors are $\pm 5\text{--}10\%$.

The Fe concentration is high, because of the possible contamination or deliberated mixing of blue pigments with hematite-based red pigments or other rocks or clays, and also the presence of Fe ions that substitute Mg and/or Al in the octahedral layer of the palygorskite [10].

A high concentration of Ca was previously related to the use of the *fresco* technique for painting the mural [5]. In this technique, a fresh plaster of slaked lime $\text{Ca}(\text{OH})_2$ is prepared and the paint is done when the plaster is wet. In the drying process of lime, it reacts with the CO_2 from air, to form a layer of calcite (CaCO_3) on top of the paint surface. This layer is impermeable and protects the fresco paint from humidity and other environmental aggressions. Moreover, in fresco murals, the pigments cannot be easily removed using, for example, wet cotton, as it usually happens for murals painted with tempera. Another striking characteristic is that fresco murals are extremely hard, effect produced by the compaction and cementation of the calcite over hundreds of years, and are difficult to remove using, for example, a hunter's knife.

In our quantitative analysis, we found that the Ca concentrations rank from 30% to 50%, which are very close to the calcium concentration in calcite (40%) (Table 1). This calcium level is an indication that the *fresco* technique was used. As a matter of fact, PIXE is very sensitive to the surface, and reveals a superficial calcite layer which protects the pigments. According to SRIM2000 computer code [11], the projected range of 3 MeV protons in a calcite matrix is 71 μm , therefore, it is expected that in our experiments we see the external calcite and the pigment layer embedded in it. These should be of the order of tens of microns. An exception is the mural sample from Santa

Cecilia Acatitlán, which gave much lower concentration of Ca, suggesting that this sample was not prepared using the fresco technique. Another sample we analyzed (not reported in Table 1) is a blue cornice from Cacaxtla, which also showed a very low Ca concentration (around 3%) meaning that is not made using the fresco technique, which is reasonable because this sample does not belong to a mural paint but to a decoration.

Copper is absent or only present at low trace concentration (below 30 $\mu\text{g/g}$). This is an important fact, indicating the absence of minerals that are typically used for blue (azurite) or green (malaquite). The same comment applies to cobalt, but with a higher concentration (below 500 $\mu\text{g/g}$). Co has not been used for the statistical analysis because the obtained values were very close or below the limit of detection.

The principal components analysis (PCA) [12] is used to identify groups with common characteristics in highly correlated systems. In our case, the interest is to check whether temporal (colonial and Prehispanic) and/or spatial (geographic) groups come from the compositional analysis of the blue pigment. In the PCA analysis the 70% or more of the total variance in a sample population can be described using only the first three principal components (PC), similarly to what was observed for pottery, ceramics [12] and obsidians.

We performed a PCA analysis using the covariance of the logarithm of the concentrations reported in Table 1. The first three principal components reproduced the 71.15% of the total variance, in agreement with a previous publication [12] (34.88, 24.95 and 11.32 for the first, second and third PCs, respectively). The plot of the first and sec-

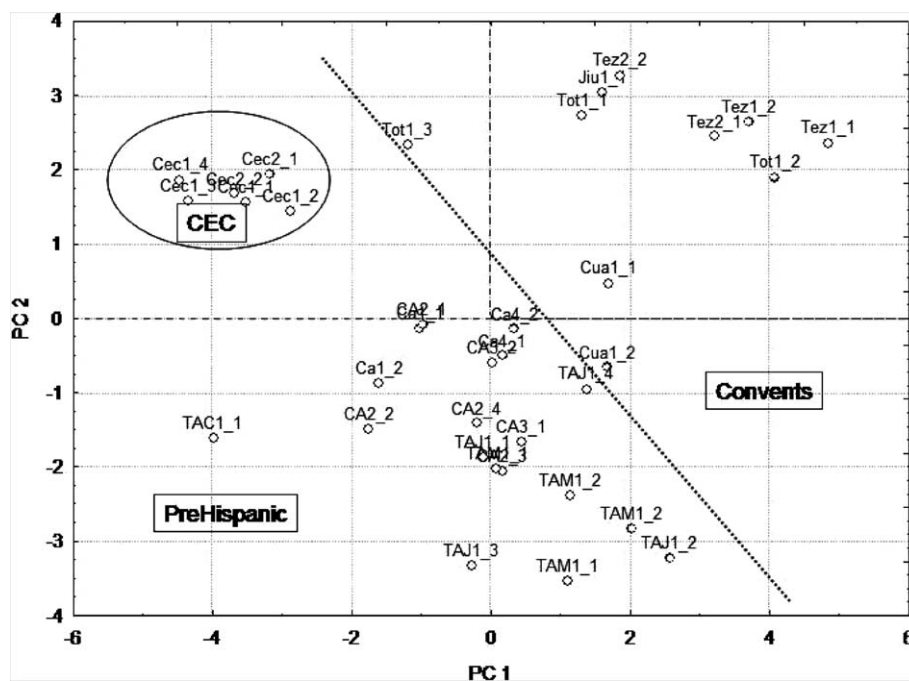


Fig. 2. PCA plot (second PC versus first PC) showing two well defined zones, corresponding to Prehispanic and colonial samples. Note also that the CEC samples, which are not made using the fresco technique, form themselves a well-defined cluster.

ond PCs (Fig. 2) shows that the plane can be separated in two zones (dashed line) one corresponding to the Prehispanic samples, and the other to colonial samples. This separation clearly identified the two groups in the most obvious historical classification. It is interesting that the samples from CEC (Santa Cecilia Acatitlan) cluster together in a small zone. This is due to the fact that these samples are not painted using the fresco technique, thus are very different from the others. If we look inside a given zone (Prehispanic and colonial), it can be observed that samples from the same sites sometimes are close together, in particular for the Prehispanic sites (like CA or TAM). In the case of the colonial samples, those coming from the same site tend to be more distant one to another. Moreover, it looks not possible to separate the Agustian convent (TEZ) from Franciscan ones (CUA, TOT and JIU). As a general comment, although the samples can be differentiated in colonial and Prehispanic groups, these groups are not well separated in the PCs planes, suggesting that both the Prehispanic and colonial techniques for mural paints were similar, probably due to the fact that colonial artists learned from Prehispanic ones, or even, indigenous artists decorated the convents, as has been documented in some cases, like in the Tecamachalco convent (Puebla).

To summarize, we studied for the first time a collection of samples with Maya blue pigment from both Prehispanic and colonial sites. Using quantitative PIXE analysis is possible to identify when a mural was painted using a technique which is not fresco if Ca concentration is very different from 40%. PCA analysis differentiated two temporal groups (Prehispanic and colonial). However, the data did not cluster into well defined groups, suggesting that

similar techniques were used in colonial times, certainly imported from the Prehispanic artists.

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References

- [1] H. Van Olphen, *Science* 154 (1966) 645.
- [2] R. Kleber, R. Masschelein-Kleiner, J. Thissen, *Stud. Conserv.* 12 (1967) 41.
- [3] C. Reyes-Valerio, De Bonampak al Templo Mayor. El azul maya en Mesoamerica, Colección America Nuestra, Vol. 40, 1993, Mexico DF: Siglo XXI editores, p. 157.
- [4] A.A. Tagle, H. Paschinger, H. Richard, G. Infante, *Stud. Conserv.* 35 (1990) 156.
- [5] M. Sanchez del Río, P. Martinetto, A. Somogyi, C. Reyes-Valerio, E. Dooryhée, N. Peltier, L. Alianelli, B. Moignard, L. Pichon, T. Calligaro, J.-C. Dran, *Spectrochim. Acta B* 59 (2004) 1619.
- [6] J.A. Maxwell, J.L. Campbell, W.J. Teesdale, *Nucl. Instr. and Meth. B* 43 (1989) 218.
- [7] J.A. Maxwell, W.J. Teesdale, J.L. Campbell, *Nucl. Instr. and Meth. B* 95 (1995) 407.
- [8] K. Govindaraju, *Geostand. Newslett.* 19 (1995) 1.
- [9] StatSoft, *Statistica*. Available from: <<http://www.statsoftinc.com/>>.
- [10] M. Sanchez del Río, A. Sodo, S. Eeckhout, T. Neisius, P. Martinetto, E. Dooryhée, C. Reyes-Valerio, *Nucl. Instr. and Meth. B* 238 (2005) 50.
- [11] J.F. Ziegler. Available from: <<http://www.srim.org/>>.
- [12] M. Glaskock, in: H. Neff (Ed.), *Chemical Characterization of Ceramic Pastes in Archaeology*, Monographs in World Archaeology, Prehistoric Press, Madison, WI, 1992.